

First Principles of Time Series Tokenizers

Research Proposal

Aysin Tumay

UC San Diego

Abstract

What problem are you trying to solve? What are your envisioned solution(s)? How does it contribute to cutting-edge research in machine learning?

1 Research Question

The central question of this project is whether LLM performance on time series tasks is determined by the tokenizer rather than the architecture. Despite the field’s focus on scaling backbones and pretraining corpora, forecasting benchmarks show that two models with identical architectures can differ substantially based solely on how inputs are discretized, yet no prior work has asked what a tokenizer must formally guarantee for an LLM to reason faithfully about time series, nor whether those guarantees transfer across qualitatively different tasks such as long-horizon forecasting and action-conditioned transition modeling in offline RL. This gap matters because tokenizer failures in clinical or control settings are not benchmark regressions but safety failures. We hypothesize that a small set of axiomatic properties are jointly necessary and sufficient for reliable LLM reasoning across both tasks, and that existing tokenizers fail in axiom-traceable ways: their downstream failure modes directly reflect which properties they violate.

2 Proposed Solution

Although there are some recent works targeting design knobs of time series forecasting with LLMs, the first principles of LLM time series tokenizers for high-stakes applications have yet to be explored. There is no single method that satisfies all principles listed below.

Let $x \in \mathbb{R}^T$ denote a univariate time series of length T , $x \in \mathbb{R}^{N \times T}$ a multivariate time series with N channels, $\mathcal{Z}$ the token space, and $\mathcal{T} : \mathbb{R}^k \rightarrow \mathcal{Z}^p$ a tokenizer of output length p operating on patches of length k . We define a patch starting at time t with length k as $\mathbf{x}_t := x_{[t:t+k)}$. Within the token set mapped by $\mathcal{T}$, there exist three functional components: $\mathcal{T}_{\text{shape}} : \mathbb{R}^k \rightarrow \mathcal{Z}_{\text{shape}}$, $\mathcal{T}_{\text{affine}} : \mathbb{R}^k \rightarrow \mathcal{Z}_{\text{affine}}$, and $\mathcal{T}_{\text{multi}} : \mathbb{R}^{N \times k} \rightarrow \mathcal{Z}_{\text{multi}}$, addressing morphological structure, scale and offset, and cross-channel co-dependency, respectively. The full token set

is the composite

$$\mathcal{T}(\mathbf{x}_t) = (\mathcal{T}_{\text{shape}}(\mathbf{x}_t), \mathcal{T}_{\text{affine}}(\mathbf{x}_t), \mathcal{T}_{\text{multi}}(\mathbf{x}_t)) \in \mathcal{Z}_{\text{shape}} \times \mathcal{Z}_{\text{affine}} \times \mathcal{Z}_{\text{multi}} = \mathcal{Z}^p, \quad (1)$$

where $\mathbf{x}_t \in \mathbb{R}^{N \times k}$ is the multivariate patch matrix across all N channels at time t when used as input to $\mathcal{T}_{\text{multi}}$. Each axiom below is stated in terms of whichever component(s) it constrains.

Axiom 1 (Affine Sensitivity). *The affine component must be injective under affine transformations of the input:*

$$\mathbf{x}_t \neq A\mathbf{x}_t + B \implies \mathcal{T}_{\text{affine}}(\mathbf{x}_t) \neq \mathcal{T}_{\text{affine}}(A\mathbf{x}_t + B) \quad \forall A \in \mathbb{R}^k, B \in \mathbb{R}^k. \quad (2)$$

$\mathcal{T}_{\text{affine}}$ must encode the full signal including its baseline and amplitude. This rules out any normalization step that discards A or B before tokenization (e.g., RevIN, instance normalization), as these enforce $\mathcal{T}_{\text{affine}}(A\mathbf{x}_t + B) = \mathcal{T}_{\text{affine}}(\mathbf{x}_t)$ by construction. Since $\mathcal{T}(\mathbf{x}_t) \neq \mathcal{T}(A\mathbf{x}_t + B)$ whenever any component differs, Axiom 1 is satisfied by $\mathcal{T}_{\text{affine}}$ alone.

Axiom 2 (Multivariate Co-dependency). *Let $\mathbf{x}_t^{(i)}$ and $\mathbf{x}_t^{(j)}$ be the i -th and j -th channel patches where $i \neq j$. The multivariate component must preserve mutual information across channels:*

$$\mathbf{x}_t^{(i)} \not\perp \mathbf{x}_t^{(j)} \implies \mathcal{T}_{\text{multi}}(\mathbf{x}_t)^{(i)} \not\perp \mathcal{T}_{\text{multi}}(\mathbf{x}_t)^{(j)}, \quad (3)$$

where $\mathcal{T}_{\text{multi}}(\mathbf{x}_t)^{(i)}$ denotes the token component corresponding to channel i . Intuitively, if covariates carry mutual information, the tokens encoding their joint context must also carry that information. This axiom is not satisfiable by $\mathcal{T}_{\text{shape}}$ or $\mathcal{T}_{\text{affine}}$ alone, as both operate per-channel independently; $\mathcal{T}_{\text{multi}}$ is the necessary component.

Axiom 3 (Phase Awareness). *Let σ_τ denote the temporal shift operator $\sigma_\tau \mathbf{x}_t = \mathbf{x}_{t-\tau}$. The shape component must be shift-variant:*

$$\sigma_\tau \mathbf{x}_t \neq \mathbf{x}_t \implies \mathcal{T}_{\text{shape}}(\sigma_\tau \mathbf{x}_t) \neq \mathcal{T}_{\text{shape}}(\mathbf{x}_t) \quad \forall \tau \neq 0. \quad (4)$$

Tokens must encode the temporal position of a pattern, not just its shape. A phase-unaware tokenizer maps $\mathcal{T}_{\text{shape}}(\sigma_\tau \mathbf{x}_t) = \mathcal{T}_{\text{shape}}(\mathbf{x}_t)$ for all τ , collapsing the same waveform at different phases to the same token and making it impossible for the downstream model to reason about when a pattern occurs.

Axiom 4 (Shape Awareness). *Define the normalized shape of a patch as $\bar{\mathbf{x}}_t = (\mathbf{x}_t - \mu_{\mathbf{x}_t})/\sigma_{\mathbf{x}_t}$. The shape component must be discriminative over normalized morphology while being invariant to affine transformation:*

$$\mathcal{T}_{\text{shape}}(\alpha \mathbf{x}_t + \beta) = \mathcal{T}_{\text{shape}}(\mathbf{x}_t) \quad \forall \alpha, \beta, \quad (5)$$

$$d(\bar{\mathbf{x}}_t, \bar{\mathbf{y}}_t) > \epsilon \implies \mathcal{T}_{\text{shape}}(\mathbf{x}_t) \neq \mathcal{T}_{\text{shape}}(\mathbf{y}_t), \quad (6)$$

where d is a morphology-sensitive distance (e.g., DTW or ℓ_2 on the normalized signal). Shape awareness and Axiom 1 are complementary by design: $\mathcal{T}_{\text{shape}}$ is affine-invariant while $\mathcal{T}_{\text{affine}}$ is affine-sensitive, so together the composite $\mathcal{T}$ satisfies both simultaneously. $\mathcal{T}_{\text{shape}}$ additionally provides a morphologically meaningful distance over the codebook.

Axiom 5 (Reconstruction Fidelity). *There must exist a decoder $\mathcal{D} : \mathcal{Z} \rightarrow \mathbb{R}^k$ such that the reconstruction error is uniformly bounded:*

$$\sup_{x \in \mathcal{X}} \|\mathcal{D}(\mathcal{T}(x)) - x\|_2 \leq \varepsilon$$

Axiom 5 is a necessary condition for using the tokenizer inside a world model where the output is a predicted next state $\hat{s}_{t+1}$. If $\mathcal{T}$ is not approximately invertible, the predicted token cannot be mapped back to a meaningful physiological observation. WaveToken [Masserano et al., 2025] satisfies this exactly via the inverse DWT (Parseval’s theorem ensures $\|\mathcal{D}(\mathcal{T}(x)) - x\|_2 \rightarrow 0$ as quantization resolution increases); TOTEM [Talukder et al., 2024] satisfies it approximately via a VQ-VAE decoder minimizing $\mathbb{E}_x[\|\mathcal{D}(\mathcal{T}(x)) - x\|_2^2]$.

2.1 Our Method

The dominant narrative in time series foundation models has centered on scaling transformer architectures and pretraining corpora, yet forecasting benchmarks reveal a quieter truth: **two models with identical architectures and training budgets can differ substantially in performance based solely on how their inputs are discretized**. Specifically, a poorly designed tokenizer forces the downstream model to solve two problems simultaneously: learn what the signal means, and learn to predict it. Separating these concerns, committing to a tokenizer that encodes physiological structure explicitly determines what the model is capable of learning.

Settling the importance of a tokenizer, using a purely preprocessing-based tokenizer or a trained architecture is a task-dependent choice that is strained by the richness of the data and the learning power of the LLM. Our clinical dataset is small by construction since collecting trajectories is expensive. Training a tokenizer from scratch on such data risks learning representations that overfit to the limited signal distribution rather than capturing generalizable physiological morphology. A pretrained tokenizer, trained on large corpora of diverse time series, provides an encoder that already organizes the latent space around recurring temporal patterns. Finetuning this tokenizer on MCS data with a reconstruction objective then adapts the codebook entries to the specific morphological vocabulary of hemodynamic and pump waveforms, without discarding the general structure learned during pretraining.

To satisfy the axioms above at the same time, here is the full pipeline formulation until LLM input.

Patch and scale extraction. For each channel $i \in \{1, \dots, N\}$ and time step t , extract a patch $\mathbf{x}_t^{(i)} \in \mathbb{R}^k$ and compute per-patch statistics:

$$\mu_t^{(i)} = \frac{1}{k} \sum_{t'=t}^{t+k} x_{t'}^{(i)}, \quad (7)$$

$$\sigma_t^{(i)} = \sqrt{\frac{1}{k} \sum_{j=t}^{t+k} \left(x_{t'}^{(i)} - \mu_t^{(i)}\right)^2} + \epsilon. \quad (8)$$

The pair $(\mu_t^{(i)}, \sigma_t^{(i)})$ is stored as a scale token and bypasses the codebook entirely.

Normalization. Each patch is normalized to unit scale before any further processing:

$$\bar{\mathbf{x}}_t^{(i)} = \frac{\mathbf{x}_t^{(i)} - \mu_t^{(i)}}{\sigma_t^{(i)}}. \quad (9)$$

All downstream stages until the decoder operate on $\bar{\mathbf{x}}_t$.

Cross-channel attention. Let $N = 12$ denote the number of channels and let $\bar{\mathbf{X}}_t \in \mathbb{R}^{C \times k}$ denote the normalized patch matrix across all channels at time t , where each row $\bar{\mathbf{x}}_t^{(i)} \in \mathbb{R}^k$ is the normalized patch of channel i . Let $W^Q, W^K, W^V \in \mathbb{R}^{k \times d}$ and $W^O \in \mathbb{R}^{d \times k}$ be learned projection matrices shared across all channels.

For each channel $i \in \{1, \dots, N\}$, compute the query vector and the key and value matrices over all channels:

$$\mathbf{q}^{(i)} = \bar{\mathbf{x}}_t^{(i)} W^Q \in \mathbb{R}^d, \quad (10)$$

$$\mathbf{K} = \bar{\mathbf{x}}_t W^K \in \mathbb{R}^{N \times d}, \quad (11)$$

$$\mathbf{V} = \bar{\mathbf{x}}_t W^V \in \mathbb{R}^{N \times d}. \quad (12)$$

The attention weight vector over all N channels is:

$$\mathbf{a}^{(i)} = \text{softmax}\left(\frac{\mathbf{q}^{(i)} \mathbf{K}^\top}{\sqrt{d}}\right) \in \mathbb{R}^N, \quad (13)$$

where the j -th entry $a^{(i,j)}$ encodes how much channel i attends to channel j . The attended context vector and its projection back to patch space are:

$$\mathbf{h}^{(i)} = \mathbf{a}^{(i)} \mathbf{V} \in \mathbb{R}^d, \quad (14)$$

$$\tilde{\mathbf{h}}^{(i)} = \mathbf{h}^{(i)} W^O \in \mathbb{R}^k. \quad (15)$$

The residual connection and layer normalization are applied in patch space, where both terms are $\in \mathbb{R}^k$:

$$\tilde{\mathbf{x}}_t^{(i)} = \text{LayerNorm}\left(\bar{\mathbf{x}}_t^{(i)} + \tilde{\mathbf{h}}^{(i)}\right). \quad (16)$$

This operation is applied independently at each time patch, yielding the cross-attended representation $\tilde{\mathbf{x}}_t \in \mathbb{R}^{N \times k}$. The output $\tilde{\mathbf{x}}_t^{(i)}$ encodes pump–hemodynamic co-dependencies (Axiom 3) while remaining on the normalized scale.

Shape proxy construction. To satisfy Axiom 4, morphological descriptors computed on the normalized signal are as follows:

$$\mathbf{F}_t^{(i)} = \left[\mu_t^{(i)}; \sigma_t^{(i)}; \mathcal{M}_{\min}\left(\tilde{\mathbf{x}}_t^{(i)}\right); \mathcal{M}_{\max}\left(\tilde{\mathbf{x}}_t^{(i)}\right) \right] \in \mathbb{R}^{k+4}, \quad (17)$$

where $\mathcal{M}_{\min}, \mathcal{M}_{\max}$ are the patch extrema. Note that $\mathbf{F}_t^{(i)}$ is *not* passed to the encoder; they are stored separately for tokenization.

Encoding and vector quantization. Remember that N denotes the number of channels and k the patch length. The pretrained VQ-VAE encoder E (borrowed from TOTEM Talukder et al. [2024]), with shared weights and frozen parameters, processes each channel patch independently, mapping the normalized patch matrix to a continuous latent matrix:

$$E : \mathbb{R}^{N \times k} \rightarrow \mathbb{R}^{N \times d}, \quad \mathbf{Z}_t = E(\tilde{\mathbf{x}}_t) \in \mathbb{R}^{N \times d}, \quad (18)$$

where d is the encoder latent dimension and the i -th row $\mathbf{z}_t^{(i)} \in \mathbb{R}^d$ is the continuous latent representation of channel i . Vector quantization maps each row of $\mathbf{z}_t$ to the nearest entry in the codebook $\mathcal{C} = \{\mathbf{c}_1, \dots, \mathbf{c}_{N_c}\} \subset \mathbb{R}^d$:

$$\text{VQ} : \mathbb{R}^{N \times d} \rightarrow \{1, \dots, N_c\}^N, \quad q_{\text{shape},t}^{(i)} = \arg \min_{n \in \{1, \dots, N_c\}} \left\| \mathbf{z}_t^{(i)} - \mathbf{c}_n \right\|_2, \quad (19)$$

where N_c is the codebook size and $q_{\text{shape},t}^{(i)} \in \{1, \dots, N_c\}$ is the discrete shape code for channel i at time t . The retrieved codebook vector $\mathbf{c}_{q_{\text{shape},t}^{(i)}} \in \mathcal{C} \subset \mathbb{R}^d$ is used for reconstruction at the decoder and in the VQ loss. The full pipeline contracts the normalized patch matrix $\tilde{\mathbf{x}}_t \in \mathbb{R}^{N \times k}$ to a vector of shape codes $\mathbf{q}_{\text{shape},t} \in \{1, \dots, N_c\}^N$, one discrete index per channel.

The scalar patch statistics are quantized independently using channel-specific bins, without passing through the encoder:

$$q_{\mu,t}^{(i)} = \text{Bin}_{\mu}^{(i)}(\mu_t^{(i)}), \quad (20)$$

$$q_{\sigma,t}^{(i)} = \text{Bin}_{\sigma}^{(i)}(\sigma_t^{(i)}), \quad (21)$$

$$q_{\min,t}^{(i)} = \text{Bin}_{\min}^{(i)}(\mathcal{M}_{\min,t}^{(i)}), \quad (22)$$

$$q_{\max,t}^{(i)} = \text{Bin}_{\max}^{(i)}(\mathcal{M}_{\max,t}^{(i)}). \quad (23)$$

The per-channel reward token for $\mathcal{R} = \{\text{MAP}, \text{HR}, \text{Pulsat}\}$ is quantized as:

$$q_{r,t}^{(i)} = \begin{cases} r_t^{(i)} + 8 \in \{1, \dots, 8\} & i \in \mathcal{R}, \\ 0 & i \notin \mathcal{R}, \end{cases} \quad (24)$$

where $r_t^{(i)} \in [-7, 0]$. The full discrete token at time t for channel i is:

$$\tau_t^{(i)} = \left(q_{\text{shape},t}^{(i)}, q_{\mu,t}^{(i)}, q_{\sigma,t}^{(i)}, q_{\min,t}^{(i)}, q_{\max,t}^{(i)}, q_{r,t}^{\text{MAP}}, q_{r,t}^{\text{HR}}, q_{r,t}^{\text{Pulsat}} \right) \in \{1, \dots, N_c\} \times \{1, \dots, B\}^4 \times \{0, \dots, B_r\}^3. \quad (25)$$

LLM world model and decoder. The world model g receives the token sequence across all channels and the action sequence, and learns the autoregressive distribution over next tokens:

$$p(\tau_{\leq t}, a_{\leq t}) = g\left(\{\tau_s^{(i)}\}_{s \leq t, i \leq N}, \{a_s\}_{s \leq t}\right), \quad (26)$$

where a_t encodes the P-level at time t . The world model predicts the next shape code $\hat{q}_{\text{shape},t+1}^{(i)} \in \{1, \dots, N_c\}$, from which the corresponding codebook vector is retrieved:

$$\hat{\mathbf{c}}_{t+1}^{(i)} = \mathbf{c}_{\hat{q}_{\text{shape},t+1}^{(i)}} \in \mathcal{C} \subset \mathbb{R}^d. \quad (27)$$

The decoder $D : \mathcal{C} \rightarrow \mathbb{R}^k$ maps the retrieved codebook vector to a normalized patch shape. Affine restoration using the predicted scale statistics recovers the physiological signal with the normalization statistics of $\mathbf{x}_t$:

$$\hat{\mathbf{x}}_{t+1}^{(i)} = \sigma_t^{(i)} \cdot D\left(\hat{\mathbf{c}}_{t+1}^{(i)}\right) + \mu_t^{(i)} \in \mathbb{R}^k. \quad (28)$$

Tokenizer finetuning. The encoder E is frozen throughout. The cross-channel attention weights (W^Q, W^K, W^V, W^O), the codebook entries $\mathcal{C}$, and the decoder D are jointly optimized on MCS data. The scalar bin boundaries for q_μ , q_σ , $q_{\min}$, $q_{\max}$, q_r are fixed from training set statistics and are not updated. The total tokenizer loss is:

$$\mathcal{L}_{\text{tokenizer}} = \mathcal{L}_{\text{VQ}} + \mathcal{L}_{\text{recon}}. \quad (29)$$

The VQ commitment loss moves codebook entries toward the frozen encoder outputs:

$$\mathcal{L}_{\text{VQ}} = \left\| \text{sg}\left[\mathbf{z}_t^{(i)}\right] - \mathbf{c}_{q_{\text{shape},t}^{(i)}} \right\|_2^2 + \beta \left\| \mathbf{z}_t^{(i)} - \text{sg}\left[\mathbf{c}_{q_{\text{shape},t}^{(i)}}\right] \right\|_2^2, \quad (30)$$

where $\text{sg}[\cdot]$ is the stop-gradient operator and $\beta > 0$ is the commitment weight. Since E is frozen, the second term regularizes the codebook entries to remain close to the encoder output distribution without updating the encoder. The reconstruction loss trains the decoder D and the cross-channel attention weights:

$$\mathcal{L}_{\text{recon}} = \text{MSE}(\mathbf{x}, \hat{\mathbf{x}}) + \lambda \text{sDTW}(\tilde{\mathbf{x}}, \hat{\hat{\mathbf{x}}}), \quad (31)$$

where $\hat{\hat{\mathbf{x}}} = D\left(\mathbf{c}_{q_{\text{shape},t}^{(i)}}\right) \in \mathbb{R}^k$ is the normalized-scale decoder output, $\tilde{\mathbf{x}}$ is the normalized ground truth patch, and $\hat{\mathbf{x}} = \sigma_t^{(i)} \cdot D\left(\mathbf{c}_{q_{\text{shape},t}^{(i)}}\right) + \mu_t^{(i)}$ is the reconstruction in raw scale. The sDTW term penalizes morphological deviation on the normalized signal while MSE operates on the denormalized space. Gradients from $\mathcal{L}_{\text{recon}}$ reach the cross-channel attention weights via the straight-through estimator, which substitutes $\mathbf{z}_t^{(i)}$ for $\mathbf{c}_{q_{\text{shape},t}^{(i)}}^{(i)}$ during backpropagation.

Lemma 1 (Fixed-Position Token Structure). *The tokenizer $\mathcal{T}$ emits a fixed-arity tuple in which each slot is occupied by exactly one component, in a fixed order independent of the input:*

$$\mathcal{T}(\mathbf{x}_t) = \left(\underbrace{\mathcal{T}_{\text{shape}}(\mathbf{x}_t)}_{\text{slot 1}}, \underbrace{\mathcal{T}_{\text{affine}}(\mathbf{x}_t)}_{\text{slot 2}}, \underbrace{\mathcal{T}_{\text{multi}}(\mathbf{x}_t)}_{\text{slot 3}} \right), \quad (32)$$

where slot m takes values in the per-slot space $\mathcal{Z}_m$ for $m \in \{\text{shape}, \text{affine}, \text{multi}\}$. Consequently the joint token space is the Cartesian product of the per-slot spaces:

$$\mathcal{Z}^p = \mathcal{Z}_{\text{shape}} \times \mathcal{Z}_{\text{affine}} \times \mathcal{Z}_{\text{multi}}. \quad (33)$$

Proof. By construction, $\mathcal{T}$ concatenates the outputs of its three components into a tuple of fixed length p whose slot assignment does not depend on $\mathbf{x}_t$: slot 1 always carries the shape code, slot 2 the affine code, and slot 3 the multivariate code. Since the slot positions are fixed and each slot ranges over its own space $\mathcal{Z}_m$ independently of the others, the set of all attainable tuples is exactly the product set $\mathcal{Z}_{\text{shape}} \times \mathcal{Z}_{\text{affine}} \times \mathcal{Z}_{\text{multi}}$. No collapsing or merging of slots into a shared index occurs, so this product structure is preserved. $\square$ $\square$

Lemma 2 (Component Independence). *Under the fixed-position structure of Lemma 1, the composite token is equal if and only if all components are simultaneously equal:*

$$\mathcal{T}(\mathbf{x}_t) = \mathcal{T}(\mathbf{y}_t) \iff \mathcal{T}_{\text{shape}}(\mathbf{x}_t) = \mathcal{T}_{\text{shape}}(\mathbf{y}_t) \wedge \mathcal{T}_{\text{affine}}(\mathbf{x}_t) = \mathcal{T}_{\text{affine}}(\mathbf{y}_t) \wedge \mathcal{T}_{\text{multi}}(\mathbf{x}_t) = \mathcal{T}_{\text{multi}}(\mathbf{y}_t). \quad (34)$$

Equivalently, the value of any one slot places no constraint on the values of the others, and a change in any single component strictly changes the tuple.

Proof. By Lemma 1, $\mathcal{T}(\mathbf{x}_t)$ and $\mathcal{T}(\mathbf{y}_t)$ are elements of the Cartesian product $\mathcal{Z}_{\text{shape}} \times \mathcal{Z}_{\text{affine}} \times \mathcal{Z}_{\text{multi}}$. Two elements of a Cartesian product are equal if and only if they agree in every coordinate; this is the defining property of equality on a product set. Applying this coordinate-wise criterion to the three slots yields the stated biconditional. In particular, if any component differs in its slot, the tuples differ regardless of the other slots, so there is no interaction between slots that could cancel an individual difference. $\square$ $\square$

Case 1: Affine statistics differ.

Suppose $(\mu_x^{(i)}, \sigma_x^{(i)}) \neq (\mu_y^{(i)}, \sigma_y^{(i)})$ for some channel i . Since $\mathcal{T}_{\text{affine}}$ encodes $(\mu_t^{(i)}, \sigma_t^{(i)})$ per channel via scalar quantization (Axiom 1), any difference in affine statistics that exceeds the bin resolution $\varepsilon_{\text{affine}}$ produces

$$\mathcal{T}_{\text{affine}}(\mathbf{x}_t) \neq \mathcal{T}_{\text{affine}}(\mathbf{y}_t). \quad (35)$$

By Lemma 1 it follows immediately that $\mathcal{T}(\mathbf{x}_t) \neq \mathcal{T}(\mathbf{y}_t)$.

Case 2: Affine statistics agree but normalized morphology differs.

Suppose $(\mu_x^{(i)}, \sigma_x^{(i)}) = (\mu_y^{(i)}, \sigma_y^{(i)})$ for all i , but $\bar{\mathbf{x}}_t^{(i)} \neq \bar{\mathbf{y}}_t^{(i)}$ for some i , where $\bar{\mathbf{x}}_t^{(i)} = (\mathbf{x}_t^{(i)} - \mu_x^{(i)})/\sigma_x^{(i)}$. Since affine statistics are identical, $\mathcal{T}_{\text{affine}}(\mathbf{x}_t) = \mathcal{T}_{\text{affine}}(\mathbf{y}_t)$. The difference is carried entirely in the normalized shape. By Axiom 4, $\mathcal{T}_{\text{shape}}$ is discriminative over normalized morphology: whenever $d(\bar{\mathbf{x}}_t^{(i)}, \bar{\mathbf{y}}_t^{(i)}) > \varepsilon_{\text{shape}}$,

$$\mathcal{T}_{\text{shape}}(\mathbf{x}_t) \neq \mathcal{T}_{\text{shape}}(\mathbf{y}_t), \quad (36)$$

and by Lemma 1, $\mathcal{T}(\mathbf{x}_t) \neq \mathcal{T}(\mathbf{y}_t)$. For the subcase $d(\bar{\mathbf{x}}_t^{(i)}, \bar{\mathbf{y}}_t^{(i)}) \leq \varepsilon_{\text{shape}}$, the two patches are morphologically indistinguishable within codebook resolution and their controlled collapse is sanctioned by Axiom 5.

Case 3: Per-channel marginals agree but cross-channel dependency differs.

Suppose $\bar{\mathbf{x}}_t^{(i)} = \bar{\mathbf{y}}_t^{(i)}$ and $(\mu_x^{(i)}, \sigma_x^{(i)}) = (\mu_y^{(i)}, \sigma_y^{(i)})$ for all channels i . Then $\mathcal{T}_{\text{affine}}(\mathbf{x}_t) = \mathcal{T}_{\text{affine}}(\mathbf{y}_t)$ and $\mathcal{T}_{\text{shape}}(\mathbf{x}_t) = \mathcal{T}_{\text{shape}}(\mathbf{y}_t)$. Since $\mathbf{x}_t \neq \mathbf{y}_t$ by assumption, there must exist a pair of channels (i, j) such that the joint structure $(\mathbf{x}_t^{(i)}, \mathbf{x}_t^{(j)})$ and $(\mathbf{y}_t^{(i)}, \mathbf{y}_t^{(j)})$ differ in their cross-channel dependency. By Axiom 2, $\mathcal{T}_{\text{multi}}$ preserves this mutual information: whenever the cross-channel dependency structure differs beyond quantization tolerance $\varepsilon_{\text{multi}}$,

$$\mathcal{T}_{\text{multi}}(\mathbf{x}_t) \neq \mathcal{T}_{\text{multi}}(\mathbf{y}_t), \quad (37)$$

and by Lemma 1, $\mathcal{T}(\mathbf{x}_t) \neq \mathcal{T}(\mathbf{y}_t)$.

Exhaustiveness and global collapse.

Cases 1–3 partition all ways in which $\mathbf{x}_t \neq \mathbf{y}_t$ can arise. In each case, at least one component produces a distinct output. By Lemma 1, a global collapse of $\mathcal{T}(\mathbf{x}_t) = \mathcal{T}(\mathbf{y}_t)$

would require all three components to be simultaneously equal. Since the three cases are exhaustive and each is handled by a distinct component, no such simultaneous equality can occur outside the controlled quantization tolerance $\varepsilon = \max(\varepsilon_{\text{shape}}, \varepsilon_{\text{affine}}, \varepsilon_{\text{multi}})$.

3 Related Work

Table 1: Coverage of tokenizer axioms across state-of-the-art time series foundation models. $\checkmark$ = fully addressed; $\sim$ = partially; $\times$ = not addressed.

Model	Affine sensitivity	Multivariate co-dep.	Phase awareness	Shape awareness	Reconstruction fidelity
PatchTST Nie et al. [2023]	$\times$	$\times$	$\times$	$\sim$	$\times$
Crossformer Zhang and Yan [2023]	$\times$	$\checkmark$	$\times$	$\sim$	$\times$
iTransformer Liu et al. [2024]	$\times$	$\checkmark$	$\times$	$\sim$	$\times$
Chronos Ansari et al. [2024]	$\sim$	$\times$	$\times$	$\times$	$\sim$
Chronos-Bolt Ansari et al. [2024]	$\checkmark$	$\times$	$\times$	$\sim$	$\times$
Moirai Woo et al. [2024]	$\times$	$\sim$	$\times$	$\sim$	$\times$
TOTEM Talukder et al. [2024]	$\times$	$\times$	$\times$	$\checkmark$	$\checkmark$
VQShape Zhang et al. [2024]	$\sim$	$\times$	$\checkmark$	$\checkmark$	$\checkmark$
WaveToken Masserano et al. [2025]	$\sim$	$\times$	$\checkmark$	$\checkmark$	$\checkmark$
CVPE Yu et al. [2025]	$\times$	$\checkmark$	$\times$	$\sim$	$\times$

The literature most relevant to our work spans three overlapping themes: patch-based and inverted Transformer architectures for multivariate time series, foundation models that learn discrete tokenized representations, and LLM-based approaches to time series modelling and model-based reinforcement learning. We survey each in turn, situating GORMPO’s design choices against the limitations exposed by this body of work.

3.1 Patch-Based and Inverted Transformer Architectures

PatchTST Nie et al. [2023]: Segments each univariate channel into fixed-width patches and applies self-attention independently per channel. Channel independence ignores the tight physiological coupling between hemodynamic variables (pump speed, LVEDP, pump pressure), and fixed patch widths impose a rigid temporal resolution that cannot adapt to event-driven ICU signals. Our tokenizer encodes cross-channel dependencies before quantization (Axiom 3).

Crossformer Zhang and Yan [2023]: Adds explicit cross-dimension attention on top of temporal patch attention, capturing inter-variate structure that PatchTST misses. However, it produces *continuous* contextual embeddings, cannot serve as the bottleneck of a discrete-token world model, and scales quadratically with the number of variates.

iTransformer Liu et al. [2024]: Treats each variate as a single token spanning the full lookback window and applies attention across variates. Inter-channel structure is well captured, but all within-channel temporal resolution is collapsed into one dense embedding, and it produces no discrete token sequence compatible with autoregressive prediction.

3.2 Foundation Models with Discrete Tokenization

Chronos Ansari et al. [2024]: Applies uniform bin quantization to mean-scaled univariate series and pre-trains a T5 encoder-decoder on the resulting discrete vocabulary. Bin boundaries are shared across all variates and samples, so signals with very different physiological ranges (pump current vs. systolic pressure) receive identical resolution. Channels are processed independently and there is no provision for control signals, making direct use in action-conditioned offline RL impossible.

Moirai Woo et al. [2024]: Uses multi-resolution patching with a mixture-distribution prediction head to achieve zero-shot forecasting across diverse domains. It outputs continuous predictive distributions rather than discrete latent tokens, contains no mechanism for action conditioning, and its closed-source training corpus limits reproducibility in clinical settings.

TOTEM Talukder et al. [2024]: Learns a domain-agnostic codebook via a convolutional VQ-VAE trained on univariate reconstruction. The local convolutional receptive field limits long-range temporal modelling, strided convolutions can destroy phase alignment (Axiom 4), and the univariate objective provides no incentive to encode cross-channel dependencies critical to hemodynamic weaning signals.

VQShape Zhang et al. [2024]: Represents subsequences as a normalized shape plus separate amplitude, scale, and offset attributes, yielding interpretable and domain-generalizable tokens. For cardiovascular hemodynamics, however, the absolute magnitude of a signal excursion carries as much clinical weight as its shape, so discarding amplitude at tokenization introduces a bottleneck for reward estimation. VQShape is also evaluated exclusively on classification benchmarks.

WaveToken Masserano et al. [2025]: Applies a fixed discrete wavelet transform per channel and threshold-quantizes the resulting coefficients before autoregressive forecasting. Three limitations apply directly to our setting: the DWT is computed independently per channel (no cross-channel interaction), the wavelet basis and threshold are fixed a priori rather than adapted per sample, and there is no control-signal conditioning.

Yu et al. [2025]: Provides a systematic empirical analysis of how design choices, such as patch size, normalization, positional embedding, induce implicit biases in time series foundation models, including a regression-to-the-mean effect from common patch-embedding schemes. Rather than proposing a method, this work motivates our axiomatic framework: Axiom 1 (Affine Sensitivity) directly guards against the mean-collapsing bias it identifies.

CVPE Shin and Zhang [2025]: CVPE injects cross-variate context into channel-independent models by inserting a router-attention block before the patch embedding step: learnable router vectors first aggregate information across all variates, then redistribute it back into per-channel patch embeddings, leaving the rest of the backbone unchanged. Integrated into Time-LLM, this yields gains on datasets with strong inter-variate correlations (Weather, Traffic) but hurts performance on weakly correlated ones (ETTh2, ETTm2), suggesting that indiscriminate channel mixing at the embedding level is sensitive to correlation structure. Two further gaps apply to our setting: CVPE produces continuous embeddings rather than discrete tokens and has no mechanism for action conditioning, so it cannot serve as a tokenizer for an offline-RL world model.

3.3 Time Series LLM Literature

Time-LLM Jin et al. [2024]: Reprograms a frozen LLM by projecting input patches onto learnable text prototypes, enabling zero-shot forecasting without backbone fine-tuning. The reprogramming module is trained on a fixed forecasting horizon, does not support step-by-step action-conditioned rollouts, and operates channel-independently.

LLMTime Gruver et al. [2023]: Serializes normalized time series as decimal strings for direct LLM next-token prediction. Byte-pair encoding splits numeric tokens inconsistently, degrading precision for signals with narrow clinically significant ranges, and each channel is queried in isolation with no notion of a control signal.

DICL Benechehab et al. [2025]: The work most directly related to ours. DICL projects multivariate trajectories into PCA-disentangled components and feeds each independently to an LLM for zero-shot transition prediction, demonstrating that pre-trained LLMs can serve as dynamics models for offline RL. Three limitations motivate our design: PCA is a linear operation that misses non-linear hemodynamic coupling; predicting each principal component in isolation discards cross-channel correlations; and the method offers no safety guarantees distinguishing in-distribution from out-of-distribution transitions. GORMPO addresses these through a discrete tokenizer that preserves multivariate structure before in-context prediction and a density-based pessimism penalty with formal offline-support guarantees.

4 Baselines

For our time series forecasting task, we will compare our method to the following baselines also mentioned in the related work section. The LLM model will kept stationary and the tokenizer will be swapped.

- **PatchTST:** Channel-independent patch Transformer that processes each variate in isolation, providing a strong CI forecasting baseline with no cross-variate structure.
- **iTransformer:** Inverted Transformer that embeds each variate as a single token and applies attention across variates, capturing inter-channel dependencies without discrete tokenization.
- **Chronos:** Pre-trained T5 encoder-decoder that tokenizes time series via uniform bin quantization, serving as a discrete but channel-agnostic and action-unaware foundation model baseline.
- **TOTEM:** Convolutional VQ-VAE that learns a domain-agnostic discrete codebook from a univariate reconstruction objective, ablating our axiomatic tokenizer design without cross-channel attention.
- **WaveToken:** Autoregressive model over threshold-quantized discrete wavelet coefficients, representing a fixed frequency-domain alternative to our learned discrete tokenization.

For the RL experiments, our baselines are as follows:

- **DICL**: Uses PCA to linearly disentangle multivariate trajectories before feeding each component independently to an LLM for zero-shot transition prediction, providing the closest prior to our LLM-based world model.
- **TOTEM + MBRL**: TOTEM’s convolutional VQ-VAE used as the transition model within our pessimistic MBRL loop, isolating the effect of axiomatic tokenizer design over an unstructured discrete alternative.
- **iTransformer + MBRL**: iTransformer adapted as a transition model by appending the pump-level action as an additional variate, providing a continuous cross-variate dynamics baseline within the same MBRL framework.
- **PatchTST + MBRL**: PatchTST adapted as a transition model by appending the pump-level action as an additional channel, serving as the simplest continuous CI dynamics baseline within our MBRL framework.
- **naive MBRL**: model-based policy optimization with no uncertainty penalty and special tokenization.

5 Experiment Plan

5.1 Datasets.

We evaluate the forecasting component of our tokenizer on five standard long-term forecasting benchmarks and one proprietary clinical dataset.

ETT (ETTh1, ETTh2, ETTm1, ETTm2) Zhou et al. [2021]. The Electricity Transformer Temperature datasets record seven variates (oil temperature and six power-load features) from two electricity transformers in China over two years. ETTh1 and ETTh2 are sampled hourly; ETTm1 and ETTm2 at 15-minute intervals. We follow the standard train/validation/test split and evaluate on prediction horizons $H \in \{96, 192, 336, 720\}$.

Weather Wu et al. [2021]. Records 21 meteorological indicators (temperature, humidity, air pressure, etc.) every 10 minutes throughout 2020, collected by the Max Planck Institute for Biogeochemistry. With 21 variates and a fine sampling rate, it probes cross-channel and high-frequency modelling capacity. Horizons: $H \in \{96, 192, 336, 720\}$.

MCS. The Mechanical Circulatory Support dataset comprises de-identified ICU records from patients supported by the Impella device. Each trajectory contains 12 hemodynamic variables (PumpPressure, PumpSpeed, PumpFlow, LVP, LVEDP, Systolic, Diastolic, PulsatilityIndex, PumpCurrent, HeartRate, ESE_{lv}, and P-level P2–P10) sampled at 10-minute intervals, with a 6-step prediction horizon aligned to the clinical decision cycle.

For the RL application, we use our proprietary clinical dataset due to its time series control setting. Evaluation will be with the Transformer-based Digital Twin. If necessary, we can include the MIMIC-4 dataset Johnson et al. [2023] due to its clinical and time series properties. The evaluation will be based on several off-policy evaluation (OPE) methods such as Importance Sampling, Doubly Robust etc.

5.2 Experiments

Here we show the main experiments to support the full pipeline along with the tokenizer axioms.

1. Verify the tokenizer + LLM if it fulfills the axioms. This could be an ablation study, but highly needed to support the soundness of our axioms empirically.
2. Test performance in time series forecasting in related datasets.
3. RL Performance with our world model as transition model compared to other world models as transition models in MBRL pipeline.

6 Evaluation Metric

Metrics for the time series forecasting task are: CRPS for uncertainty calibration, MSE, and MAE for accuracy. For the MCS forecasting task, we will also include Trend Accuracy, and MSE, MAE on the static and dynamic P-level test samples.

Metrics for the RL task are Reward, Action Change Penalty and Weaning Score in the MCS task. If MIMIC-4 is adopted later, or we want to support our model-based OPE evaluation in MCS, following OPE methods will be utilized: Weighted Importance Sampling Precup et al. [2000], Fitted Q-evaluation Le et al. [2019], Doubly Robust Jiang and Li [2016].

To rule out the effect of RL model, and the main LLM model in the final performance, tokenizer evaluation methods will be researched.

We will use open source datasets except for the MCS data. We will publish the codebase for the tokenizer along with the whole pipeline.

7 Bottlenecks and Backup Plans

Think of potential bottlenecks of your project and suggest alternative solutions if your initial idea does not work out.

1. We essentially propose to represent 6×12 time series points with 8×12 tokens. This could **badly affect token fertility, consequently model efficiency**. One solution could be to assess the performance gain attained by each token component and eliminate some if the fertility cost is higher than the performance gain. Especially, since our training objective includes softDTW which supports shape preservation, q_{min} , q_{max} could be ditched.
2. Following the previous analysis of informativeness of each token, we might include slope terms instead of q_{min} , q_{max} if the **trend accuracy of the LLM is worse** than the baselines.
3. **Freezing the encoder weights of a pretrained VQ-VAE model could create misalignment** with the cross-attended inputs since TOTEM Talukder et al. [2024] uses raw normalized inputs to the encoder model. Our motivation to freeze the weights

was not to overfit to our small MCS dataset. To mitigate this, we can train the encoder for the large time series datasets (weather and ETT). For MCS, we can concatenate our dataset with a similar hemodynamic dataset to stay in-domain. Then, we would tackle overfitting the TOTEM model. This also mitigates overfitting due to domain specificity.

4. **Performance differences between models trained on MCS versus larger datasets** could stem from dataset size rather than patch diversity. If the results are aligned with this conclusion, we will subsample the ETT and weather datasets and move on with those versions.
5. **Glitch tokens** are mixed in the inference time when some typos in text frequently appear in the corpus. This could be related to time series when the signal is noisy and the tokenizer overfits to noise which is directly routed to LLM inference. To overcome this,
6. Overfitting could appear in different ways which we touched on, such as glitch tokens, high out-of-vocabulary rate, high unknown token probability etc. Capping the vocabulary size could help in keep less probable tokens out of the vocabulary.

8 Timeline

Break down your project into milestones and elaborate on the timeline to complete them.

- **June 28:** The full pipeline is ready with a naive tokenizer and LLM head for forecasting task.
- **July 3:** Tokenizer and forecaster evaluation is implemented. MCS, ETT, Weather data loaders are implemented and work with the naive tokenizer.
- **July 10:** Chronos and WaveToken are implemented and tested on all datasets for the forecasting task.
- **July 17:** TOTEM is implemented and tested on all datasets for the forecasting task. World model implemented in MBRL pipeline.
- **July 24:** PatchTST and iTransformer are implemented and tested on all datasets for the forecasting task.
- **July 31:** Our method is implemented, trained and tested on all datasets.
- **Aug 7:** Train/test Chronos and WaveToken in the RL task on all datasets. Test our method’s cross-channel attention sensitivity and multivariate co-dependency axiom. Iterate over this property if our method is not responsive for this.
- **Aug 14:** Train/test TOTEM in the RL task on all datasets. Test our method in shape-awareness axiom by iterating on the shape proxy component and assess the finetuning objective.

- **Aug 21:** Train/test PatchTST and iTransformer in the RL task on all datasets. Test our method’s vector quantization quality. Iterate if not high quality - evaluate the tokenizer on reconstruction quality.
- **Aug 28:** Finalize baseline experiments on all tasks (including RL baselines). Train/test our method on all datasets for forecasting task.
- **Sep 4:** Train/test our method on all datasets for the RL task.
- **Sep 11:** Train/test the remaining datasets. Iterate on our method and develop ablation studies for testing our ability to fulfill the axioms.
- **Sep 18:** Writing starts and remaining experiments continue. **Abstract deadline.**
- **Sep 25:** Writing. **Paper deadline.**

References

- Abdul Fatir Ansari, Lorenzo Stella, Caner Turkmen, Xiyuan Zhang, Pedro Mercado, Huibin Shen, Oleksandr Shchur, Syama Sundar Rangapuram, Sebastian Pineda Arango, Shubham Kapoor, Jasper Zschiegner, Danielle C. Maddix, Michael W. Mahoney, Kari Torkkola, Andrew Gordon Wilson, Michael Bohlke-Schneider, and Yuyang Wang. Chronos: Learning the language of time series. *Transactions on Machine Learning Research*, 2024. ISSN 2835-8856. URL <https://openreview.net/forum?id=gerNCVqqtR>.
- Abdelhakim Benechehab, Youssef Attia El Hili, Ambroise Odonnat, Oussama Zekri, Albert Thomas, Giuseppe Paolo, Maurizio Filippone, Ievgen Redko, and Balázs Kégl. Zero-shot model-based reinforcement learning using large language models. In *International Conference on Learning Representations*, 2025. URL <https://openreview.net/forum?id=uZFXpPrwSh>.
- Nate Gruver, Marc Finzi, Shikai Qiu, and Andrew Gordon Wilson. Large language models are zero-shot time series forecasters. In *Advances in Neural Information Processing Systems*, volume 36, 2023. URL <https://arxiv.org/abs/2310.07820>.
- Nan Jiang and Lihong Li. Doubly robust off-policy value evaluation for reinforcement learning. In *Proceedings of the 33rd International Conference on Machine Learning*, pages 652–661, 2016.
- Ming Jin, Shiyu Wang, Lintao Ma, Zhixuan Chu, James Y. Zhang, Xiaoming Shi, Pin-Yu Chen, Yuxuan Liang, Yuan-Fang Li, Shirui Pan, and Qingsong Wen. Time-LLM: Time series forecasting by reprogramming large language models. In *International Conference on Learning Representations*, 2024. URL <https://openreview.net/forum?id=Unb5CVPtae>.
- Alistair E W Johnson, Lucas Bulgarelli, Lu Shen, Alvin Gayles, Zach Shahn, Steven Horng, Mihaela van der Schaar, Tom J Pollard, Benjamin Moody, Roger G Mark, Omar Badawi, and Leo Anthony Celi. MIMIC-IV, a freely accessible electronic health record dataset. *Scientific Data*, 10(1):1, 2023.

- Hoang Le, Cameron Voloshin, and Yisong Yue. Batch policy learning under constraints. In *Proceedings of the 36th International Conference on Machine Learning*, pages 3703–3712, 2019.
- Yong Liu, Tengge Hu, Haotong Li, Haixu Wu, Shu Wang, Lintao Ma, Jianmin Wang, and Mingsheng Long. iTransformer: Inverted transformers are effective for time series forecasting. In *International Conference on Learning Representations*, 2024. URL <https://openreview.net/forum?id=JePfAI8fah>.
- Luca Masserano, Abdul Fatir Ansari, Boran Han, Xiyuan Zhang, Christos Faloutsos, Michael W. Mahoney, Andrew Gordon Wilson, Youngsuk Park, Syama Sundar Rangapuram, Danielle C. Maddix, and Yuyang Wang. Enhancing foundation models for time series forecasting via wavelet-based tokenization. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pages 43248–43275. PMLR, 2025. URL <https://proceedings.mlr.press/v267/masserano25a.html>.
- Yuqi Nie, Nam H. Nguyen, Phanwadee Sinthong, and Jayant Kalagnanam. A time series is worth 64 words: Long-term forecasting with transformers. In *International Conference on Learning Representations*, 2023. URL <https://openreview.net/forum?id=Jbdc0vT0col>.
- Doina Precup, Richard S Sutton, and Satinder P Singh. Eligibility traces for off-policy policy evaluation. In *Proceedings of the 17th International Conference on Machine Learning*, pages 759–766, 2000.
- Donghua Shin and Edwin Zhang. Enhancing channel-independent time series forecasting via cross-variate patch embedding. *arXiv preprint arXiv:2505.12761*, 2025.
- Sabera Talukder, Yisong Yue, and Georgia Gkioxari. TOTEM: Tokenized time series embeddings for general time series analysis. *Transactions on Machine Learning Research*, 2024. ISSN 2835-8856.
- Gerald Woo, Chenghao Liu, Akshat Kumar, Caiming Xiong, Silvio Savarese, and Doyen Sahoo. Unified training of universal time series forecasting transformers. In *Proceedings of the 41st International Conference on Machine Learning*, Proceedings of Machine Learning Research. PMLR, 2024. URL <https://arxiv.org/abs/2402.02592>.
- Haixu Wu, Jiehui Xu, Jialiang Wang, and Mingsheng Long. Autoformer: Decomposition transformers with auto-correlation for long-term series forecasting. volume 34, pages 22419–22430, 2021.
- Annan Yu, Danielle C. Maddix, Boran Han, Xiyuan Zhang, Abdul Fatir Ansari, Oleksandr Shchur, Christos Faloutsos, Andrew Gordon Wilson, Michael W. Mahoney, and Yuyang Wang. Understanding the implicit biases of design choices for time series foundation models. *arXiv preprint arXiv:2510.19236*, 2025. URL <https://arxiv.org/abs/2510.19236>.

- Xiyuan Zhang, Danielle C. Maddix, Abdul Fatir Ansari, Boran Han, Oleksandr Shchur, others, and Yuyang Wang. VQShape: Per-sample shape quantization for distinguishable and interpretable time series representations. In *Advances in Neural Information Processing Systems*, 2024.
- Yunhao Zhang and Junchi Yan. Crossformer: Transformer utilizing cross-dimension dependency for multivariate time series forecasting. In *International Conference on Learning Representations*, 2023. URL <https://openreview.net/forum?id=vSVLM2j9eie>.
- Haoyi Zhou, Shanghang Zhang, Jieqi Peng, Shuai Zhang, Jianxin Li, Hui Xiong, and Wancai Zhang. Informer: Beyond efficient transformer for long sequence time-series forecasting. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 35, pages 11106–11115, 2021.